

Research Methodology in CS

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Meaning and Significance

- Definition of a Research Problem
- Importance in Computer Science
- Real-world examples

Sources and Identification

- Origin of Research Problems
- Literature, gaps, curiosity
- Problem formulation techniques

Criteria & Characteristics

- Good research problem attributes
- Common errors in selection

Scope, Objectives, and Approaches

- Defining objectives
- Investigative approaches
- Data collection and instrumentation

Dialectic in Computing Research

- Argument models and logical reasoning
- Proof methods: Demonstration, Empirical, Mathematical

Deduction and Induction

- Theory building approaches
- Examples in algorithms and software engineering

- Role of modelling in CS
- Performance evaluation

Literature Review Techniques

- Effective searching and organizing
- Review matrix

Plagiarism and Ethics

- Types of plagiarism
- Avoiding unethical practices

Technical and Report Writing

- Style and clarity
- Formatting and citations

Proposal Development

- Structure of a research proposal
- Presentation and review process

- Patents, Copyrights, Designs
- Innovation to protection

International Scenario

- Global patenting process
- PCT and WIPO overview

Patent Rights & Licensing

- Technology transfer
- Licensing models

New IPR Developments

- IPR in software
- Educational institutions and case studies

References

- Stuart Melville and Wayne Goddard, *Research Methodology*
- R. Ganesan, *Research Methodology for Engineers*
- Steven Skiena, *Algorithm Design Manual*

Thank You

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